

SMARTBRIDGE

Let's Bridge the Gap

SKILL WALLET

A SmartBridge Product

Performance Testing

EduGenie: AI-Powered Personalized Learning Platform

Field	Details
Team ID	UNM123-IDMAS
Project Name	EduGenie: AI-Powered Personalized Learning Platform
Maximum Marks	5 Marks
Testing Tool	Google Lighthouse, Chrome DevTools and k6
Testing Type	Load Testing (normal and peak) and Spike Testing

Step 2: Test Scenarios

S.No	Scenario	Virtual Users	Duration	Expected Outcome
1	Baseline Login/Home page load	1	30 sec	Under 2 seconds; no errors
2	Normal EduGenie page load	10	60 sec	Avg. response under 2 sec; error < 1%
3	Peak learning-page load	50	60 sec	Avg. under 2 sec; max under 5 sec
4	Spike test	100	30 sec	Pages remain available; max under 5 sec

Step 3: Testing Code / Evidence

Sample k6 script used/planned for EduGenie load testing. Replace the URL with the actual deployed EduGenie URL before execution.

EduGenie – k6 Performance Test

```
import http from "k6/http";
import { check, sleep } from "k6";

export const options = {
  vus: 10,
  duration: "60s",
};

export default function () {
  const res = http.get("https://your-edugenie-url.com/");

  check(res, {
    "status is 200": (r) => r.status === 200,
    "response < 2s": (r) => r.timings.duration < 2000,
  });

  sleep(1);
}
```

Note: This is a sample testing script for documentation. Actual Pass/Fail values should be entered after running the test.

Step 4: Performance Test Results

Metric	Target	Actual Value	Status
Average Response Time	< 2 seconds	To be filled	To be filled
Maximum Response Time	< 5 seconds	To be filled	To be filled
Throughput	To be measured	To be filled	To be filled
Error Rate	< 1%	To be filled	To be filled
CPU Utilization	< 80%	To be filled	To be filled
Memory Utilization	< 80%	To be filled	To be filled

Step 5: Observations & Analysis

- Key Findings: To be completed after test execution.
- Bottlenecks Identified: To be completed after test execution.
- Optimization Steps Taken: To be completed after test execution.